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Review Article

A review of research on urban parking prediction

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HIGHLIGHTS

- Analyzing collaborative networks in parking prediction field: exploring focus areas and research trends.
- Summarizing characteristics, advantages, and limitations of different categories of prediction models.
- Reviewing application cases of various models in the parking prediction domain.
- Synthesizing current research status of different prediction problems and proposing future research directions.

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ABSTRACT

The rapid growth of urban traffic has intensified daily congestion, affecting both traffic flow and parking. Accurate parking prediction plays a vital role in effectively managing limited parking resources and is essential for the successful implementation of advanced intelligent systems. In an effort to comprehensively assess the latest developments in parking prediction, we curated a dataset of 639 articles spanning from 2010 to the present, using the Scopus database. Initially, we performed a bibliometric analysis utilizing VOSviewer software. These findings not only illuminate emerging trends within the parking prediction field but also provide strategic guidance for its progression. Subsequently, we categorized advancements in three focal areas: behavior prediction, demand prediction, and parking space prediction. A comprehensive overview of the present research status and future directions was then provided. The findings underscore the substantial progress achieved in current parking prediction models, achieved through diverse avenues like multi-source data integration, multi-variable feature extraction, nonlinear relationship modeling, deep learning techniques application, and ensemble model utilization. These innovative endeavors have not only pushed the theoretical boundaries of parking prediction but also significantly heightened the precision and applicability of predictive models in practical scenarios. Prospective research should explore avenues such as processing unstructured parking datasets, developing predictive models for small-scale data, mitigating noise interference in parking data, and harnessing potent platform fusion techniques. This study's significance transcends guiding and catalyzing advancement in academic and practical domains; it holds paramount relevance across academic research, technological innovation, decision-making support, business applications, and policy formulation.

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1. Introduction

In metropolitan areas worldwide, parking has consistently posed a prominent challenge, sparking growing interest in parking space guidance and search research. As urbanization gains momentum and land use becomes more concentrated, the parking dilemma intensifies. Long queues and circling for parking spaces further worsen traffic congestion on city roads during peak hours (Shoup, 2006). Low-speed vehicles operate with heightened power consumption, emissions, and noise levels, emitting a substantial volume of harmful gases that imperil human well-being, pollute urban surroundings, and yield profound repercussions for both individuals and society.

At present, parking challenges manifest in the following two main aspects. Firstly, there exists a growing parking demand while the supply of parking services remains limited. According to data from the Traffic Management Bureau of the Chinese Ministry of Public Security, the total number of motor vehicles in China reached 417 million by the end of 2022. In major Chinese cities, the ratio of private cars to available parking spaces is as low as 0.8:1, significantly trailing the ratio of 1.3:1 observed in developed cities. Data from the 2022 Urban Parking Expo in China reveals an annual demand for parking spaces around 30 million since 2016, yet the actual annual increase remains below 1 million, falling far short of market demand. Currently, a glaring deficit of 80 million parking spaces underscores the substantial gap between parking availability and demand. This pressing issue, seen in cities like Chongqing and Shanghai where approximately 30% of traffic congestion results from parking shortages, echoes similar situations in New York City, Paris, London, and Tokyo where traffic congestion is linked to insufficient parking availability. Based on statistical data from the U.S. Department of Transportation, approximately 40% of daily traffic congestion in New York City is attributed to the search for parking spaces. Similarly, reports from the European Commission indicate a strong correlation between the traffic congestion issues in Paris and London and the shortage of parking spaces, with around 25% of vehicular congestion being attributed to insufficient parking availability. Furthermore, data from the Japanese Ministry of Land, Infrastructure, Transport and Tourism reveals that approximately 70% of road traffic congestion in Tokyo is directly linked to the search for parking spaces. These facts highlight the global challenge of parking scarcity, significantly impacting urban traffic and sustainability. Secondly, car parking space utilization efficiency remains notably low. According to the 2022 Smart Parking Industry Report by the Institute of Intelligent Parking in China, over 90% of urban parking spaces are utilized at less than 50% capacity. The exacerbation of parking shortages is driven by irrational urban planning, limited land resources, inadequate parking management policies, and the continuous growth in vehicle numbers, all contributing to suboptimal parking

utilization. Imbalances between supply and demand result in prolonged searches for parking, irrational parking behavior, and parking resource wastage. Inadequate parking availability may lead to vehicles parking in inappropriate locations, exacerbating resource misallocation. Such circumstances significantly reduce parking space utilization efficiency. Hence, addressing parking shortages and improving parking resource utilization have become central challenges in contemporary urban management.

Mobile connectivity enables the comprehensive management of parking facilities, offering real-time information to travelers regarding parking lot locations, space guidance, availability, payment details, and even future space availability (Teodorović and Lucic, 2006). Advancements in communication and cloud computing technologies, coupled with enhanced infrastructure intelligence, have progressively enhanced parking facilities. Consequently, urban parking guidance systems have transitioned from conceptual notions to practical realities (Fahim et al., 2021). Intelligent parking systems present a pragmatic approach for optimizing parking efficiency (Lin et al., 2017). Within these intelligent frameworks, time-sensitive parking predictions assume a pivotal role in both driver decisions and urban planning (Caicedo, 2009). Research by Caicedo et al. (2006) indicated that drivers equipped with parking information exhibited a 45% higher parking success rate compared to those without such insights. Furthermore, parking predictions contribute to urban traffic management and planning (Barriga et al., 2019). Municipal traffic authorities can employ predicted data to regulate traffic demand and alleviate congestion. Parking operators and facility managers can anticipate system performance, enabling strategic long-term and short-term interventions to enhance operational efficiency. Additionally, parking prediction aids in reducing traffic congestion and environmental pollution. A survey revealed that U.S. drivers spend an average of 17 h annually searching for parking, incurring a per-driver cost of \$345 (Cookson and Pishue, 2017). Thus, precise parking prediction minimizes time and effort for drivers and reduces energy consumption. This factor not only holds undeniable significance within the transportation system but also poses a substantial challenge for management departments.

Due to the cyclical and stochastic nature of parking data, its patterns can vary between weekdays and weekends, influenced by factors such as transportation facility improvements, urban planning, travel demand, inclement weather, and supply. This multifaceted nature makes accurate prediction challenging. Additionally, gathering parking data involves temporal and spatial aspects, adding complexity to generating precise forecasts that incorporate both dimensions. Parking demand inherently contains relevant spatio-temporal features linked to traffic flows, events, and weather conditions. Thus, developing a model capturing the

spatio-temporal correlations among various data sources becomes a pivotal objective in forecasting car parking occupancy. Unfortunately, the effective reduction of noise interference in parking sample data is often overlooked, although noise can distort series features or obscure underlying trends, significantly affecting model convergence speed. In the context of managing parking resources effectively, prediction serves as a cornerstone. Establishing a precise and efficient prediction model remains intricate yet essential.

Recognizing the significance of forecasting within intelligent transport systems, extensive research efforts have been dedicated to this field. The advancement of intelligent transport strategies and the augmentation of computing capabilities have led to the emergence of diverse and innovative optimization models. In light of this context, the present paper employs a bibliometric approach to systematically analyze prominent literature from the past decade. This analysis aims to provide a comprehensive overview of the current research landscape, thereby furnishing theoretical foundations for the progression of intelligent transportation systems. More specifically, this study employs a survey of recent research developments to elucidate prevailing research focal points and subsequently explores prospective trends. The contributions of this study can be encapsulated as follows.

- Analyzing collaborative networks in parking prediction field: exploring focus areas and research trends.
- Summarizing characteristics, advantages, and limitations of different categories of prediction models.
- Reviewing application cases of various models in the parking prediction domain.
- Synthesizing current research status of different prediction problems and proposing future research directions.

2. Research methods

The literature is primarily sourced from the Scopus database, chosen for its comprehensive coverage and detailed search capabilities encompassing citation data, titles, abstracts, keywords, funding details, and more. This extensive scope facilitates a comprehensive econometric analysis of the relevant literature. The search strategy focuses on titles, keywords, and abstracts. Initially, the keyword parking prediction was employed to initiate the search, followed by the selection of pertinent works in fields like berth share prediction, vacant berth prediction, berth availability prediction, parking demand prediction, and parking behavior prediction. Employing this retrieval strategy yielded a total of 639 pertinent articles spanning from January 2010 to November 2022.

Visualizing the structure, distribution, and trends of scientific knowledge is facilitated by the VOSviewer software, which aids in depicting the current status and trajectories of research within a specific subject or domain over a defined time frame. The software utilizes overlay, density, and network visualizations based on node size, density, and distance to evaluate research directions and identify hotspots within the field. Furthermore, the acquired reference information was exported to VOSviewer for subsequent

econometric analysis. Through the examination of literature details (such as authors, sources, citations, and types), the relevance of titles, abstracts, and keywords was assessed. Finally, a meticulous systematic review was conducted, employing various classifications for enhanced comprehension.

3. Results of metrological analysis

3.1. Author, institution, national co-occurrence analysis

A total of 2172 authors were identified across the 639 documents obtained. We restricted the analyzed authors to publish at least two papers. Finally, 244 authors were selected for analysis. The scholarly community demonstrates a robust collaboration in the realm of parking forecasts, as illustrated in Fig. 1. Upon excluding instances of identical initials, the author who emerged as the most prolific is Jiazao Lin, affiliated with the Department of Information Management at Peking University and the Institute of Automation, Chinese Academy of Sciences. Leading a dedicated research team, Jiazao Lin has made substantial contributions in the fields of artificial intelligence, data analytics, and the internet of things over the course of several years. Their research endeavors have primarily centered around the study of parking behavior, leading to the publication of 7 scholarly papers. These publications, spanning from 2018 to the present, effectively showcase their profound insights and notable contributions within the realm of parking behavior research. This timeline suggests a current emphasis on this area of study. Following closely is Lei Peng, presently affiliated with the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences. Lei Peng's research pursuits encompass the internet of vehicles and big data mining, with a specific interest in predictive modeling for available parking spaces.

The collective affiliations of researchers reveal that the authors of the 639 publications hailed from a diverse array of 1536 institutions. The quantity of articles attributed to an institution serves as an indicator of its academic influence. To examine the evolution of research interests among distinct institutions over time, we also stipulated a minimum publication threshold of two, ultimately identifying and analyzing 87 organizations. A visual representation of this data in Fig. 2 illustrates that Chinese institutions have substantially dominated the academic landscape since 2016, particularly universities and research establishments. These institutions have played a pivotal role in fostering the advancement and technological enhancement of smart transportation solutions. Among these institutions, Southeast University leads the ranks with 21 published papers, closely followed by Tongji University with 18. Tsinghua University, Chang'an University, and the Chinese Academy of Sciences have contributed 15, 14, and 11 articles respectively. Additionally, Wuhan University, Kyoto University, Tianjin University, the University of Washington, the University of California, and Eudemin University have notably embarked on earlier research endeavors in the realm of parking forecasting,

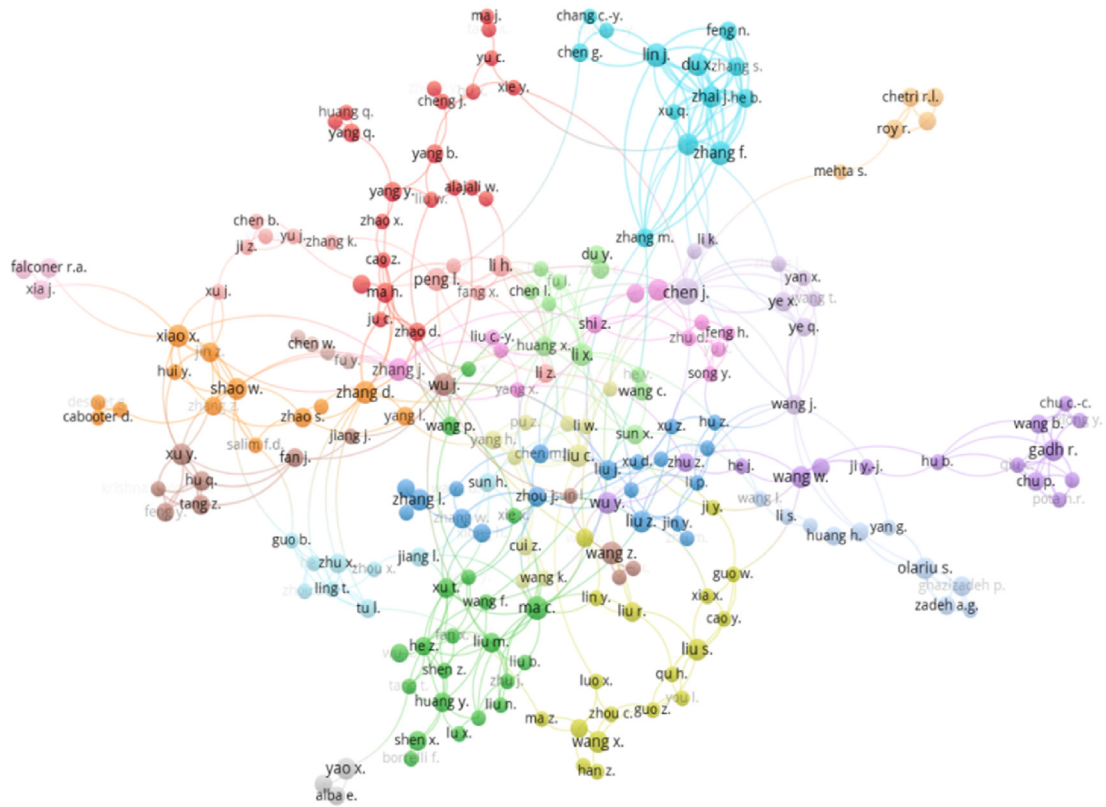


Fig. 1 – Author co-occurrence results.

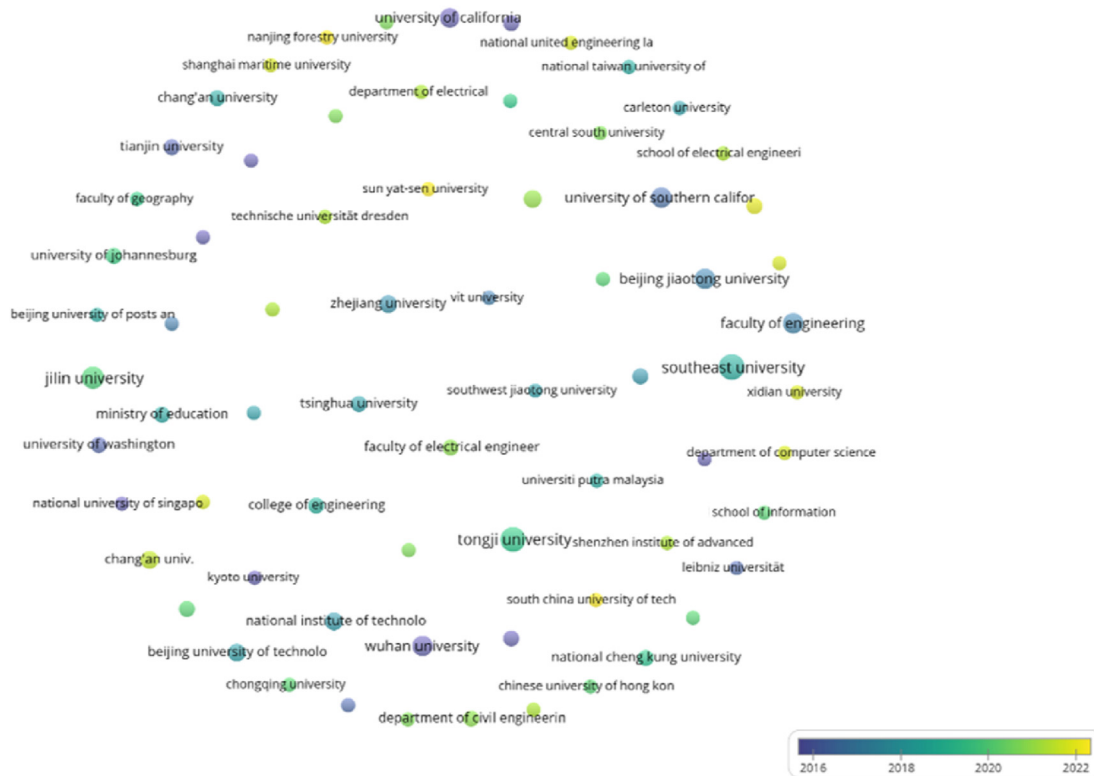


Fig. 2 – Institution co-occurrence results.

imparting a significant guiding influence in the establishment of a knowledge bridge.

Based on the authors of the papers, we conducted a further analysis of the collaborative relationships among the countries where the papers were published. By employing a minimum threshold of two papers, we examined collaborations among 41 countries and regions, as depicted in Fig. 3. Based on the number of articles published, the leading three countries and regions were China, USA, and India, contributing 256, 128, and 77 publications respectively. Moreover, China has established extensive collaborations with nations such as the United States and the United Kingdom, forming numerous cooperative partnerships. Despite China's relatively recent engagement in parking prediction research, it has successfully established a robust framework and research system for parking prediction. Through strategic initiatives, establishment of key laboratories, and collaborations with universities, China has yielded substantial outcomes, demonstrating significant progress in this field.

3.2. Keywords co-occurrence analysis

The frequency of keywords can partly indicate the research popularity and focus within a specific field. To present the outcomes more comprehensively, we set a criterion of at least eight occurrences for keyword inclusion, thereby filtering out less significant terms. Ultimately, 178 keywords were selected for analysis, and the linkage results are displayed in Fig. 4. Predominantly, high-frequency terms extracted from the literature encompassed “parking spaces”, “traffic congestion”, “intelligent systems”, “smart city”, “prediction accuracy”, “long short-term memory”, “machine learning”, “availability”, and others. The visual analysis has elucidated that the amalgamation of diverse prediction methods predominantly aims to tackle several challenges, such as environmental pollution, on-street parking facilities, and congestion in surrounding road networks. Consequently,

this offers a robust basis for the advancement of intelligent transportation systems and underscores that parking prediction is not only a trending research area but also a crucial application direction in the realm of smart parking. Notably, since 2020, machine learning and deep learning techniques have been extensively integrated into parking prediction due to their superior performance, with decision trees and long short-term memory networks emerging as the most prevalent application approaches.

Leveraging modern technology and data analysis, parking prediction methods have been implemented. This approach, through real-time monitoring of parking lots and traffic conditions, combines historical data with model algorithms to deliver accurate parking space information and guidance to users, offering a highly promising opportunity for achieving more efficient parking management. In general, the performance of parking prediction and the construction of the framework have been basically completed in the world frontier, and the research and development of new tools, new methods and new standards has become a hot topic.

3.3. Conclusions and development trends

The issue of traffic congestion has become prevalent on a global scale, serving as one of the driving forces behind the emergence of intelligent parking systems and other solutions. With ongoing technological advancements, researchers are actively exploring key technological domains such as vehicle perception, multifaceted data fusion, data analysis and prediction, and automation. Numerous researchers have made a transition from demonstrating the necessity of parking prediction in the early stage to exploring how to realize its positive effects, and from clarifying scientific concepts and objectives to concrete applications. Leading research institutions are primarily concentrated within major universities, with a central focus on advancing more intelligent and efficient parking services. This dedication to research is steadily deepening within the

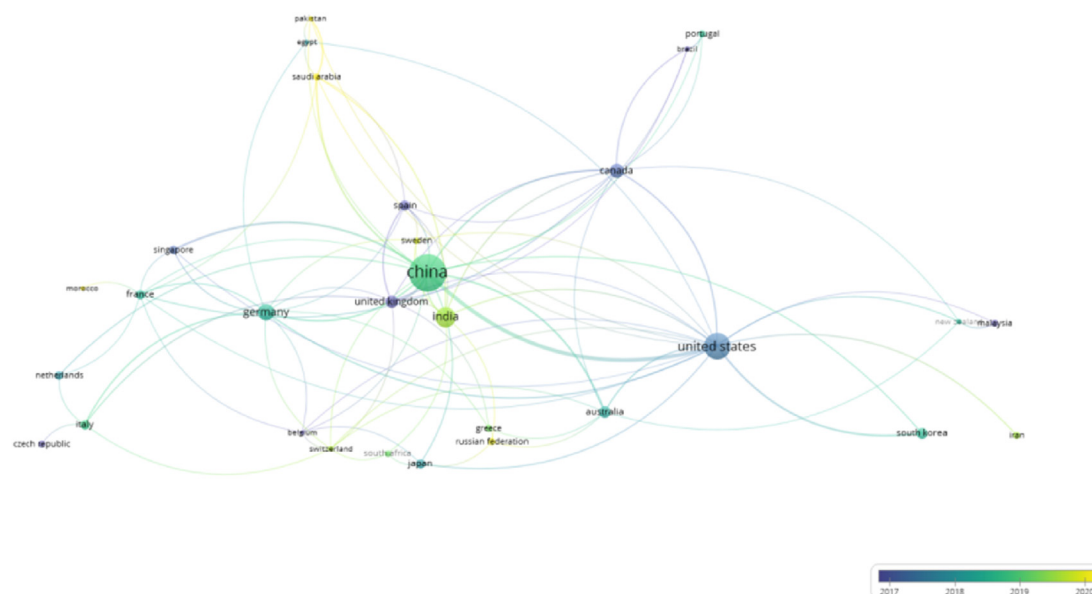


Fig. 3 – National co-occurrence results.

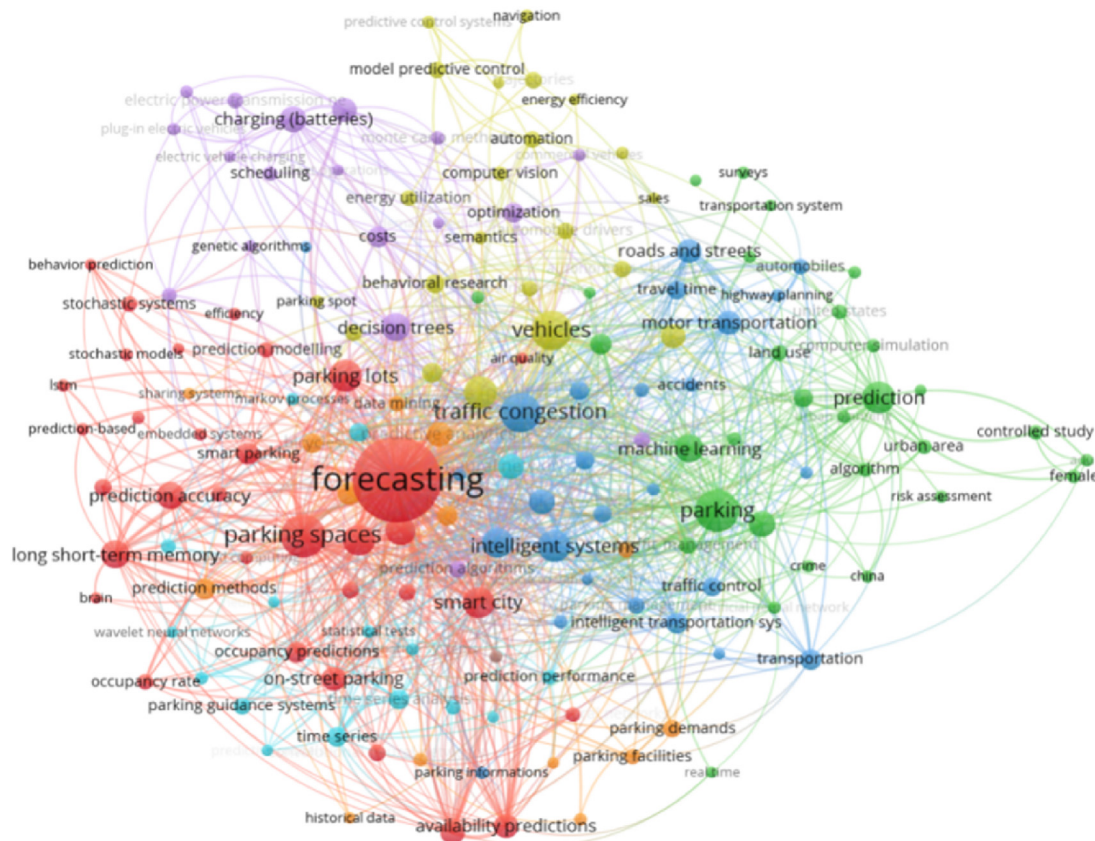


Fig. 4 – Keyword co-occurrence results.

academic community. Since 2018, China's research in the parking domain has exhibited an increasingly widespread trend, with research endeavors progressively delving deeper. Collaborative efforts between academic researchers, research institutions, and cross-border collaborations have provided robust support for the thriving development of the parking industry. The field of parking prediction is currently undergoing a phase of continuous growth and innovation. Numerous cities and parking management entities are actively employing advanced technologies and data analytics methods to enhance the accuracy and reliability of parking predictions. For instance, the comprehensive digitization of urban parking management and the implementation of intelligent technologies are gradually addressing parking challenges. These technologies systematically analyze supply-demand characteristics, parking behaviors, and preferences to scientifically plan parking needs and supply intensity, all while facilitating functionalities such as reservations. Furthermore, the integration of artificial intelligence technologies has further heightened the precision and real-time capabilities of parking prediction. Despite current challenges, such as demand fluctuations and data privacy concerns, the field of parking prediction is moving forward in a positive trajectory. The future research directions will focus on the following key areas.

- (1) Establishing regional information platforms to integrate diverse parking temporal and spatial resources, providing a solid foundation for predictive models.

- (2) Deepening the integration of parking prediction with other intelligent systems and urban infrastructure, establishing more closely linked intelligent parking systems in transportation hubs and large commercial areas.
- (3) Incorporating prediction into flexible transportation parking management, encompassing elastic transportation parking management and traffic facility management, transitioning from a static to a dynamic framework.
- (4) Holistically considering the parking demands of emerging transportation modes such as shared cars, autonomous vehicles, and electric vehicle charging stations, effectively integrating parking locations and requirements across different modes.
- (5) With the advancement of autonomous driving technology, new opportunities arise for achieving spatial-temporal balance and regulation of parking through autonomous prediction, recognition, and navigation.
- (6) With an increasing volume of data in parking prediction, a heightened emphasis on data security and privacy preservation is imperative.

These research directions not only hold the promise of bringing about new breakthroughs in the field of parking prediction but also provide robust support for urban traffic management and planning, better meeting the growing parking demands.

4. Prediction models

In this section, we will provide a detailed introduction to various predictive models, elucidating the evolution and interrelationships among different types of models. Additionally, we will conduct an in-depth comparison of classical models, exploring the strengths and limitations of each model, and drawing corresponding conclusions. On time series analysis and predicting recorded historical information, one of the earliest techniques for parking prediction in the future was developed. A time series was defined as a collection of observations, where each record represents a distinct moment in time. It is made to detect temporal trends in previously collected parking data so that the findings may be applied to forecasting in the future. Time series models include nonlinear regression, smoothing method, moving average model (MA), autoregressive moving average model (ARMA), autoregressive integrated moving average model (ARIMA), etc. With the increasing volume of parking data, data-driven transmission and prediction models are changing radically, and it is necessary to design intelligent technologies to monitor parking data. Machine learning methods have emerged accordingly, typical models include support vector machine (SVM), random forest (RF), decision tree (DT), artificial neural network (ANN) and so on. And due to the random, uncertain and non-linear nature of the parking data, machine learning methods are preferred by researchers.

Compared with the traditional machine learning techniques, the recent emerged deep learning provides enhanced performances for the parking prediction. Neural networks with deep architectures have superior learning rates than shallow networks and can be extended to high-dimensional data and support multivariate scenarios. Deep learning first converts the acquired original data into the input form, then divides the training set and the test set, trains the model and constantly adjusts the parameters until the evaluation parameter is reduced to an ideal threshold. Finally, examine the test data and evaluate the predicted results. In addition, there are two types of deep learning techniques, supervised and unsupervised, including convolutional neural network (CNN), long short-term memory network (LSTM), gated

recurrent unit (GRU), deep belief network (DBN), etc. Fig. 5 illustrates the developmental relationship of different models as the application scenarios change.

4.1. Time series models

The ARIMA model is a time series analysis method that combines autoregressive (AR), integrated (I), and MA components. In parking prediction, fixed seasonal patterns often exist, such as differences in vehicle flow between weekdays and weekends. The ARIMA model is adept at capturing these periodic fluctuations effectively (Alquraish et al., 2021). Particularly when dealing with limited dataset sizes or requiring short-term parking forecasts, the ARIMA model emerges as a practical and viable choice. Through historical data analysis and trend forecasting, the ARIMA model provides valuable predictive insights into changing parking demands. The general representation of the model is shown in Eq. (1).

$$Y_t = c + \varphi_1 Y_{t-1} + \cdots + \varphi_p Y_{t-p} + \theta_1 e_{t-1} + \cdots + \theta_q e_{t-q} + e_t \quad (1)$$

where c represents the constant term, φ_1 through φ_p correspond to the parameters of the AR model, θ_1 through θ_q correspond to the parameters of the MA model, and e_t represents the error term.

The Markov model (MM) is a probabilistic transition-based modeling approach known for its simple structure and parameters (Qi and Ishak, 2014). In the domain of parking prediction, vehicle sequences can be conceptualized as an evolution of states, and given the definition of state transition probabilities, the Markov model can effectively forecast parking conditions for the subsequent time step. Furthermore, Xiao et al. (2018) introduced a Markov model forecasting framework based on queuing models. This framework not only exhibits impressive computational efficiency but also outperforms certain machine learning methods in terms of performance.

4.2. Machine learning models

The decision-making process of DT can be visually represented as a tree-like structure, enabling DT models to provide

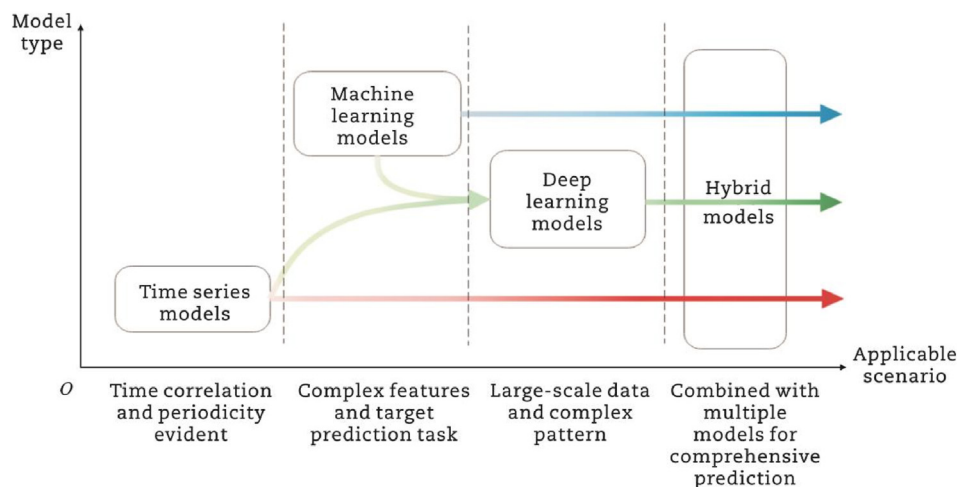


Fig. 5 – Developmental relationships of different models.

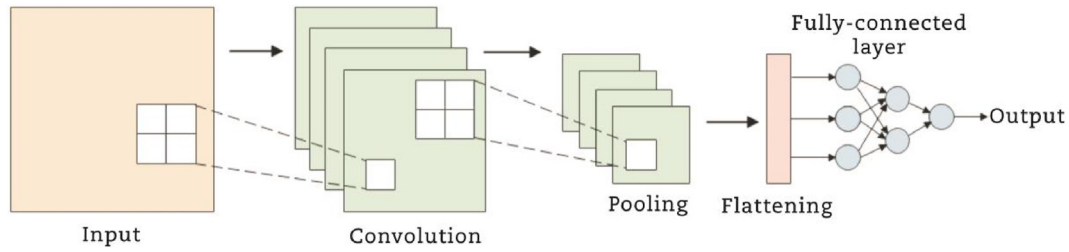


Fig. 6 – Diagram of CNN network structure (Xiao et al., 2023).

detailed explanations and reasoning processes in parking prediction. In parking prediction tasks, the DT model comprehensively considers both continuous features such as time and date, and discrete features like weekdays and holidays, forming a comprehensive and robust DT model. By evaluating the importance of each feature, the DT model can identify the most influential features on prediction outcomes, accurately selecting the required features for modeling (Zheng et al., 2016). This step assists in filtering out the most closely related features to prediction in parking prediction, thereby enhancing the predictive performance of the model. In a study by Feng et al. (2019) that addressed parking behavior prediction under weather conditions, a comparison was made among linear regression, lasso regression, DT, and RF. The study found that the DT model performed remarkably well, achieving an accuracy of 89%, further showcasing the superior performance of DT in parking prediction.

SVM exhibits superior classification performance in high-dimensional spaces, making it a potent tool for effectively handling parking data characterized by a multitude of features. Within the realm of parking prediction, SVM demonstrates the capability to map features with potential impacts on parking situations, such as time, date, weather, and other data, into high-dimensional space, thereby enabling more precise classification and forecasting. SVM not only facilitates the prioritization of feature importance through parameters like support vector weights, distances, and slack variables but also offers clear elucidation of the varying degrees of influence each feature wields in parking prediction, thus yielding results of enhanced interpretability (Mei et al., 2020). Zhao and Zhang (2020) established a parking occupancy prediction model adopting four models: linear regression, SVM, back propagation neural network (BPNN) and ARIMA. It is found that SVM can provide stable and reliable predictive capabilities for virtually all types and sizes of car parks. The model expression for SVM can be expressed as Eq. (2).

$$f(x) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i K(x_i, x) + b \right) \quad (2)$$

where α_i represents a parameter of the SVM, y_i is the category label, $K(x_i, x)$ signifies the Kernel function, and b corresponds to the bias term.

4.3. Deep learning models

CNN is a class of deep learning models extensively employed in computer vision tasks (Aladem and Rawashdeh, 2020). In

the domain of parking prediction, CNN has demonstrated the capacity to extract crucial spatiotemporal features from parking data, including variations in vehicular traffic patterns and spatial distribution of parking areas. Given the non-linear and spatially correlated nature of vehicular traffic fluctuations, the convolutional layers within CNN effectively capture local correlations and patterns within the data, while pooling layers facilitate dimensionality reduction and abstraction of local features. This enables a more precise characterization of intricate patterns present in parking data (Liu et al., 2022). The architecture of the CNN, depicted in Fig. 6, expedites model training and prediction through parallel processing, making it well-suited for real-time parking prediction scenarios with stringent timeliness requirements. This CNN-based approach introduces a novel perspective and efficacious toolset for parking prediction, with the potential to further enhance prediction accuracy and applicability.

In the context of parking prediction tasks, there often exists a significant temporal dependency, where past vehicular traffic and parking conditions exert a substantial influence on future forecasts. LSTM has showcased remarkable performance within this context. LSTM introduces an advantageous architecture by incorporating a cell state c_t alongside the original hidden layer h_t and embedding three gate mechanisms akin to switches within the cell state c_t (Ling et al., 2021). This innovative design empowers LSTM to adeptly capture and sustain long-term dependencies, thereby harnessing historical sequential information effectively for parking prediction. The fundamental structure of LSTM, depicted in Fig. 7, is capable of accommodating input sequences of

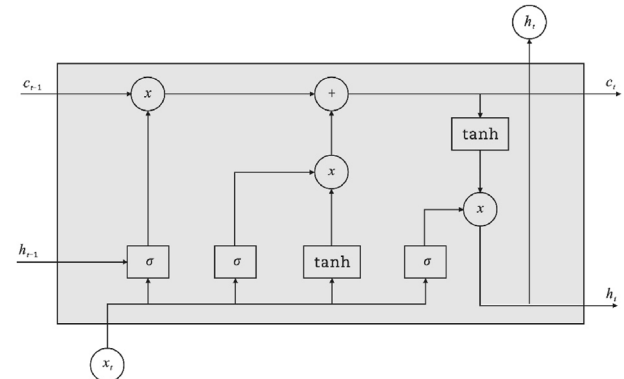


Fig. 7 – LSTM structural unit (Zeng et al., 2022b).

varying lengths and generating predictions of corresponding length. This feature enables LSTM to process historical parking data of differing durations, producing pertinent predictions at each time step, and thus rendering it suitable for real-time parking prediction applications. The integration of LSTM models presents a potent tool for addressing the challenge of long-term dependencies in parking prediction.

The computation of LSTM is listed in Eqs. (3)–(8).

$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i) \quad (3)$$

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f) \quad (4)$$

$$o_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o) \quad (5)$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (6)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (7)$$

$$h_t = o_t \odot \tanh(c_t) \quad (8)$$

where i_t , f_t and o_t signify the activation vectors associated with the input gate, forget gate, and output gate, respectively, x_t and h_t represent the input and hidden states at the current time-step, W and b represent the weight matrix and bias vector, pivotal elements undergoing learning within the network, $\tilde{c}_t$ and c_t represent the intermediate state and the resultant output state of the cellular unit, σ_g embodies the sigmoid activation function.

Deep learning, renowned for its robustness, boasts a remarkably adaptable model architecture, facilitating the tailored design and optimization of network structures based on specific problem characteristics. This inherent flexibility positions deep learning as a pervasive tool across diverse domains, encompassing the realm of parking prediction. For instance, CNN has strong spatial modelling capability (Gkoulas and Vlahogianni, 2018), LSTM may acquire features from time series separated by lengthy periods of time based on gating structure (Canli and Toklu, 2011), GRU has also become a research hotspot with the advantage of simple structure running efficiently, and DBN model is able to extract random features without manual feature extraction, the attention mechanism is extensively employed in deep learning to focus on significant features, thereby enhancing the adaptability of deep learning models in addressing the complexities of parking prediction tasks. It is often hard for a single model to predict accurate results. Hybrid modelling techniques become even more important for improving prediction results. For example, complete spatio-temporal modeling can be accomplished with CNN and LSTM (Liu et al., 2022), and the combination of LSTM and GRU (Zeng et al., 2022b) can combine the advantages of prediction accuracy and prediction efficiency.

In summary, various predictive models of different types exhibit distinct characteristics and advantages. In Tables 1 and 2, we provide a summarized overview of the merits and limitations associated with classical models and their respective categories.

5. Parking prediction

The choice of parking prediction models is often influenced by factors such as data type, data scale, prediction objectives, and specific contexts. In this section, we delve into the applications of various models within three prominent research directions: parking behavior prediction, parking demand prediction, and parking space prediction. This comprehensive overview aims to provide a detailed insight into the utilization of different models across these focal areas.

5.1. Parking behavior prediction

Parking behavior refers to the actions and habits associated with parking vehicles. Effectively understanding and predicting parking behavior can provide relevant authorities with insights into traffic conditions, enabling rational dispatching and optimization of parking resources. Parking behavior can be classified into categories such as residential parking, commuter parking, and non-commuter parking, depending on the purpose of the trip. Previous studies have analyzed parking behavior across various types and sizes of parking lots in different settings, including central urban areas (Fulman et al., 2020), commercial districts (Chiara et al., 2020), and residential neighborhoods. These parking facilities encompass public parking lots as well as shared parking spaces. The prediction models and variables used in analyzing parking behavior vary based on the specific focus of the study. Notable indicators include the number of vehicles in a parking lot and the frequency of parking events. Numerous factors influence parking behavior, such as the location of the parking facility (Ibeas et al., 2018; Zong et al., 2019), driver characteristics (Van der Waerden et al., 2015), parking rates (Frank et al., 2008; Simicevic et al., 2013), parking duration (Zong et al., 2019), walking distances (Frank et al., 2008), among others. Additionally, Feng et al. (2019) underscored the impact of weather as a crucial determinant of parking behavior and conducted an analysis that accounted for weather conditions. Furthermore, Kang et al. (2022) explored the potential influence of autonomous driving on parking behavior, while Mehta et al. (2020) investigated parking behavior in unforeseen events like the COVID-19 pandemic. Proactively predicting parking behavior can partially mitigate the challenges of finding available parking spaces, particularly during peak traffic periods.

Based on an individual user's parking history, Lin et al. (2018, 2019) introduced an intelligent parking allocation algorithm (SPA) that predicts the parking behavior of each user. Through a comparison of various models, Feng et al. (2019) determined that the RF model could accurately predict parking behavior, achieving a testing accuracy of 89%. Ling et al. (2021) utilized LSTM for prediction, leveraging multiple datasets as inputs to enhance predictive performance. This model exhibited a forecast precision of 91%. Addressing the issue of low prediction accuracy attributed to periodic parking patterns, Zhang et al. (2021) introduced the periodic weather-aware LSTM model (PewLSTM) in 2020. This model effectively anticipates parking behavior by considering

Table 1 – Strengths and limitations of classical predictive models.

Model name	Strength	Limitation
ARIMA	(1) Capture temporal trends, seasonality, and periodic patterns. (2) Simple structure, strong interpretation. (3) Predict using historical data, no need for complex features.	(1) Require adherence to stationarity presumption. (2) Challenge capturing intricate patterns in non-linear, non-stationary, noisy data. (3) Difficulty in dealing with outliers.
MM	(1) Reveal potential patterns and transfer relationships between parking data. (2) Incorporate short-term dependencies. (3) Estimate simple parameters from historical data's statistical attributes.	(1) Neglect long dependencies. (2) Sensitive to the setting of the initial state. (3) Constant transition probabilities may poorly capture actual parking dynamics.
DT	(1) The rules are clear and user-friendly, adeptly handling complex nonlinear features. (2) Key features can be automatically selected and sorted according to their importance. (3) Adapt to different types and shapes of data.	(1) Prone to overfitting, over-reliance on noise and outliers in training samples. (2) Unstable, sensitive to subtle changes in the dataset. (3) Inadequate variable correlation capture. (4) Sensitive to outliers.
SVM	(1) Suitable for high-dimensional features parking prediction. (2) Robust in the face of small amounts of noise and anomalous samples. (3) Utilize kernel functions for non-linear relationships. (4) Ability to handle parking prediction problems with small samples.	(1) When handling extensive datasets, notably Kernel function computations, entail substantial computational expenses. (2) Parameter selection affects model perform. (3) Not suitable for heavily noisy data.
CNN	(1) Can extract regional and hierarchical features. (2) Tolerates some noise and distortion in the input data. (3) Learn progressively from local to global features.	(1) High data demand. (2) Require significant computational resources and time for training. (3) Lack of interpretability of results. (4) Weak generalisation in the presence of a few class samples.
LSTM	(1) Ability to model long sequence dependencies. (2) Simultaneously capture patterns across various time scales. (3) Addresses gradient issues. (4) Robust against noise and outliers.	(1) High parameter count, causing computational complexity. (2) May face long sequence handling limitations. (3) Involve complex parameter tuning. (4) Abstract, less interpretable feature representation.

Table 2 – Strengths and limitations of the three types of prediction models.

Classification	Strength	Limitation
Time series model	(1) Capture temporal correlations, trends, seasonality, and periodic variations in parking data. (2) Utilize historical data for interpolation and imputation.	(1) Lack nonlinear relationship modeling. (2) Challenge with outliers. (3) Limited spatial correlation modeling.
Machine learning model	(1) Adapt to complex nonlinear relationships, with automated updates and robust generalization. (2) Independently extracts and selects suitable features.	(1) Poor quality, lack of diversity, or presence of noise in parking data can affect model performance. (2) Supervised learning models require a large amount of labelled parking data for training.
Deep learning model	(1) Learn high-level abstract features. (2) Can integrate multimodal data. (3) Ability to work with data in noisy environments, learn and model complex non-linear relationships. (4) Have a highly flexible model architecture.	(1) Requires a large amount of training data. (2) Large-scale models require substantial computational resources. (3) Often treated as black-box models, challenging to interpret. (4) Risk of overfitting and reduced generalization with small datasets.

historical data, weekdays, and weather conditions. Model performance was validated using real data from ten parking lots. In addition, in 2021, the same author proposed an enhanced *PewLSTM* model incorporating an event mechanism that accounts for the periodic nature of parking data and weather variations. The system overview is depicted in Fig. 8. Notably, one of China's leading smart

parking platforms, *ThsParking*, has integrated this model into its actual parking system (Zhang et al., 2021). In this study, parking behavior is delineated by the concepts of arrivals and departures. The sequence of arrivals within the past K time steps is denoted as $C = \{c_{t-K}, \dots, c_{t-2}, c_{t-1}\} \in N^K$, while the sequence of departures within the past K time steps is represented as $D = \{d_{t-K}, \dots, d_{t-2}, d_{t-1}\} \in N^K$. In this

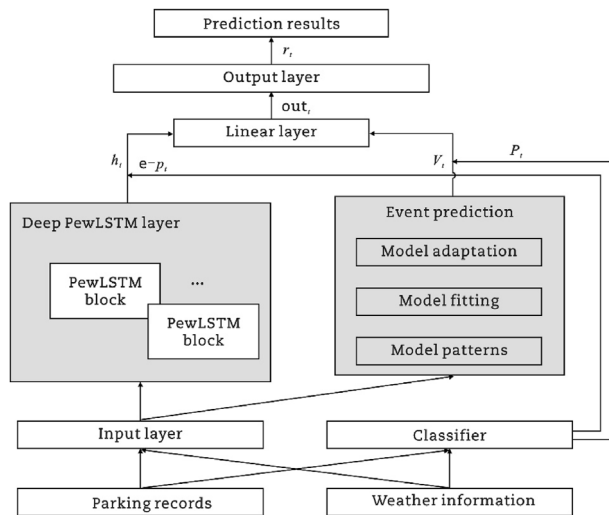


Fig. 8 – System overview of the PewLSTM (Zhang et al., 2021).

context, t signifies the current time, c_i captures the count of parking arrivals in the i th hour of the past, and d_i corresponds to the count of departures in the i th hour of the past.

At the University at Buffalo North Campus, Guo et al. (2013) introduced both a static game theory model and a dynamic neoadditive capacity model for forecasting parking behavior within a cluster of parking lots. Their research indicated that parking choice models should encompass not only objective factors such as parking occupancy status and walking distance but also consider drivers' psychological traits, as these factors significantly influence parking decisions. The incorporation of psychological characteristics into parking choice behavior models was shown to enhance predictive accuracy. In a related study, Shen et al. (2020) delved into the prediction of drivers' parking intention and maneuverability. They utilized a hierarchical LSTM model

for prediction. Table 3 specifies the factors selected for the different parking behaviour prediction models and the model characteristics.

The aforementioned findings collectively provide insights gleaned from diverse parking records, various parking types, and a range of influencing factors as studied by researchers. Key challenges in predicting parking behavior, such as the identification of influencing factors, model selection, and construction, have been effectively addressed, resulting in satisfactory prediction accuracy. These advancements contribute to the enhanced utilization of parking spaces.

It is important to note that different parking behaviors exhibit significant variations across diverse parking environments, and assessing the model's applicability necessitates careful consideration. Postpaid parking, as a prominent method for parking space sharing, embodies an intelligent and user-friendly service that underscores the efficient and streamlined management of urban parking. In the realm of predicting payment behavior for shared parking, Xu et al. (2020, 2022) have proposed a DT model and a temporal relational graph convolutional network (TR-GCN). These models have achieved accuracy rates of 79% and 91.2% respectively. Notably, the TR-GCN model entered into a partnership with the ThsParking platform for an in-depth study of payment prediction.

5.2. Parking demand prediction

The disparity between parking supply and demand often leads to the inefficient utilization of parking resources. This mismatch not only results in wasted parking spaces but also hinders the development of a sustainable transportation system. Consequently, the core objective of parking demand management lies in the precise estimation of parking requirements (Ma et al., 2023). Parking demand prediction encompasses a range of factors, including traditional elements like land usage intensity, vehicle ownership rates, parking expenses, individual travel behaviors, as well as

Table 3 – Classification and characteristics of parking behaviour prediction.

Model	Selection factor	Model feature	Literature
Structural equation model	Parking period, parking location, parking duration	Explore singled-bidirectional relationships between three parking decisions	Zong et al. (2019)
Random forest	Year, month, season, holiday, weather	Examine the impact of weather factors on parking behavior	Feng et al. (2019)
LSTM	Departure time, weather, holiday	Explain the impact of weather and holidays on parking behavior	Ling et al. (2021); Zhang et al. (2020a)
PewLSTM	Parking records, weather, environment	Has weather awareness and time prediction capability	Zhang et al. (2021)
Static game theoretic model and dynamic neo-additive capacity model	Walking distance, distance of parking spaces, psychological characteristics of individual drivers, condition of available parking spaces	Integrate the psychological characteristics of individual drivers under uncertainty	Guo et al. (2013)
Hierarchical LSTM	Forward and reverse parking maneuvers	Support for intent and multi-modal trajectory predictions	Shen et al. (2020)
PBF-SPA	Historical parking records	Predict the length of parking for each vehicle	Lin et al. (2018, 2019)

emerging influences such as public transit conditions, economic dynamics, and the integration of innovative technologies. The process of forecasting parking demand typically initiates by considering individual influences and subsequently integrating a variety of factors. On a larger scale, macro-level forecasting methods encompass diverse approaches such as travel attraction models, parking generation rate models, traffic volume-based parking demand models, and predictive models utilizing multiple regression analysis. Moreover, some models incorporate sharing theories to predict demand for shared parking, augmenting public shared parking systems by integrating demand prediction mechanisms.

As parking surveys and demand analyses have become more prevalent, parking demand forecasting models have gradually incorporated simplified survey methods and various influencing factors. These methods and factors encompass traffic demand allocation (Tong et al., 2004), parking behavior (Dave et al., 2019; Jiang et al., 2011), parking zones (Wu and Liu, 2004), and constraints related to road network capacity (Bai et al., 2004). With the advancements in data acquisition and storage technologies, parking surveys are now conducted on increasingly finer scales, and the continuous collection of parking data has become more feasible. However, the lack of information management systems in many parking facilities poses challenges in accurately identifying parking demands, whether for a small-scale or large-scale urban context. To address this, several scholars have proposed various methods for identifying parking demand (Dong et al., 2021; Zakhem and Smith-Colin, 2021; Zan et al., 2022), aiming to support efficient urban parking demand perception, prediction, and management. In terms of existing work, parking demand forecasting can be categorized into short-term and long-term forecasts. The word “long-term” refers to macro-level demand forecasts conducted over years, primarily used to shape parking policies and planning. On the other hand, “short-term” forecasting focuses on changes in intra-day parking demand, encompassing both immediate and future parking needs. Demand prediction utilized in intelligent parking guidance systems also falls under the category of short-term parking demand prediction (Boyles et al., 2015). Building upon significant existing research, this paper classifies parking demand forecasting methods into traditional approaches and machine learning techniques.

Within the domain of traditional learning methods, a subset of these strategies aims to unravel the patterns associated with parking arrivals and departures, subsequently translating them into statistical formulations for forecasting parking demand. Leveraging the prevalence of common parking entry and exit patterns, Zheng et al. (2020) crafted a parking demand prediction model rooted in the Markov birth and death process. The model's parameters underwent calibration through curve fitting and the application of uncertainty coefficient methods. Concurrently, a noteworthy traditional forecasting model is the parking generation rate model. This model's fundamental premise entails regarding diverse land parcels within the study area as sources of parking attraction, with the collective impact of these individual parcels reflecting the overall parking demand of the region (Eq. (9)). While

this approach is both intuitive and practical, the necessity of a comprehensive and accurate parking survey remains pivotal to ensure the precision of parking demand predictions. Han and Zhang (2021) introduced a behavior-based parking generation rate model, achieving a 24.45% reduction in required parking spaces compared to the conventional model. Cheng et al. (2012) adapted the generation rate model by integrating road network capacity factors, encompassing variables such as parking turnover rate, parking space occupancy rate, service level, and parking charges (Eq. (10)). Wu and Fan (2011) modified the travel attraction model by incorporating regional development factors and applied it across two distinct Chinese regions. This model posits that land parcel characteristics shape the extent of the parking attraction rate. Al-Sahili and Hamadneh (2016) proposed a parking generation rate model encompassing 73 land use categories. Furthermore, Xie et al. (2014) introduced a time-period land structure assessment model, extending the model's applicability across diverse time dimensions while also integrating various parking land and timing scenarios. Multiple regression analysis is employed to account for the combined influences of multiple factors-ranging from population and economics to land use and motor vehicle ownership-on parking demand. The selection of contributing factors for this model exerts a significant impact on the precision of estimation outcomes. Ibeas et al. (2011) demonstrated that incorporating spatial variability led to more accurate predictions when employing multiple linear regression (MLR) and geographically weighted regression (GWR) models in areas featuring paid short-term parking systems. Meanwhile, Ou and Yang (2011) devised a parking demand forecasting model grounded in the traffic flow generated by project inducements. In summary, the domain of traditional parking demand forecasting methods encompasses a diverse array of models, each tailored to capture various aspects of parking behavior. These models leverage factors such as land use, occupancy rates, and regional characteristics to provide insights into parking demand patterns. While they exhibit practicality, they often require comprehensive surveys and data collection to ensure their accuracy and applicability to real-world scenarios.

$$y = \sum_{i=1}^n a_i R_i \quad i = 1, 2, \dots, n \quad (9)$$

$$y = \left(\sum_{i=1}^n \frac{a_i R_i}{\mu_i \gamma_i} \right) \delta L \beta \quad i = 1, 2, \dots, n \quad (10)$$

where y denotes the parking demand within the central cultural zone, a_i refers to the quantities of parking demand per unit area, R_i represents the individual area measured in square meters (m^2), μ_i stands for the parking turnover rates, γ_i represents the utilization rate of parking, δ signifies the parking service level, L and β assume roles as coefficients responsible for quantifying the influences of parking prices and vehicle growth, respectively.

Machine learning plays a pivotal role in parking demand prediction, with neural network models being a central focus.

The accuracy of neural network-based parking demand predictions is substantially influenced by the network's architecture, a facet that has spurred fresh avenues of scholarly investigation. [Tang et al. \(2022\)](#) devised an innovative approach by amalgamating multiple regression and back propagation (BP) neural network models to anticipate parking demand. Network parameters were optimized using the beetle antennae search (BAS) algorithm, enhancing the algorithm's overall robustness. [Guan et al. \(2020\)](#) explored the interplay between traffic attributes, land use, and parking demand. They harnessed SVM to extract parking demand, accurately estimating it with origin-destination (OD) data from the traffic system. In a similar vein, [Long et al. \(2010\)](#) contended that diverse factors, encompassing economic, land, and traffic attributes, exert considerable influence on parking demand. Principal component analysis was employed to distill these influential variables, followed by the application of the BP neural network to predict parking demand. [Liu et al. \(2021\)](#) addressed the prediction of inflow and outflow within shared parking systems by crafting a predictive model composed of three modules: CNN, ConvLSTM, and LSTM. [Chou et al. \(2011\)](#) harnessed a variety of models, including SVM, recurrent neural networks (RNN), and LSTM, for demand forecasting. Spatio-temporal dependencies in parking demand were a focal point for [Xiao et al. \(2021b\)](#), who exploited spatio-temporal graph convolutional networks (STGCN) to anticipate demand across multi-view data. [Han \(2022\)](#) employed extreme gradient boosting (XGBoost) and DT to forecast shared parking demand. For short-term demand estimation in on-street carparks, [Liu et al. \(2022\)](#) devised a method centered on CNN and LSTM. This approach effectively harnessed

spatio-temporal characteristics to enhance generality in predictions. Advancing this paradigm, [Li and Zhong \(2023\)](#) integrated GRU with graph convolutional network (GCN), achieving more precise predictions by merging historical demand patterns with spatio-temporal traits. [Li and Zhong \(2023\)](#) introduced a novel multi-scale perspective by incorporating signal decomposition and deep learning algorithms. This approach markedly streamlined sequence complexity and non-linearity. A synthesis of common models and their key attributes across both approaches is summarized in [Table 4](#). The study of parking demand prediction within the framework of machine learning has expanded to include various models, each addressing distinct aspects of prediction accuracy, spatio-temporal dependencies, and data fusion strategies. This demonstrates the diverse and evolving landscape of machine learning's application in enhancing parking demand forecasting.

An examination of the aforementioned studies underscores the diverse array of factors that shape drivers' parking behavior, which in turn impacts parking demand. These factors can be broadly classified into socio-economic, parking policy, and parking facility-related categories. Existing models exhibit a trend of incorporating an extensive range of data sources, encompassing temporal-spatial characteristics that exert an influence on parking demand. The aim is to enhance model inputs by integrating an array of influencing factors, thereby achieving more refined forecasting outcomes. Moreover, more intricate composite models have demonstrated their robust predictive capacities by synergistically amalgamating multiple algorithms and layering multi-tier neural networks. This approach underscores the growing recognition of the nuanced interplay of various influences on

Table 4 – Classification and characteristics of parking demand prediction.

Classification of method	Model	Object of prediction	Characteristic	Duration	Literature
Traditional methods	Markov process	University parking lot	Explain the changing nature of parking demand	Short term	Zheng et al. (2020)
	Improved parking generation rate model	Urban shared parking	Simulate parking preferences, walking distances, etc.	Short term	Han and Zhang (2021)
	Improved parking generation rate model	Central business district	Consider the occupancy rate of parking space, parking rate, service level, etc.	Short term	Cheng et al. (2012)
	Multiple regression model	Areas with paid short-term parking systems	Fitting ability is strong and can be used in spatially heterogeneous situations	Short term	Ibeas et al. (2011)
Machine learning methods	Multiple regression-BP	Urban office commercial complex	Fit the nonlinear part of parking demand	–	Tang et al. (2022)
	SVM, RNN, MLP, LSTM	Central urban area, urban shared parking	Low error rate of prediction results for recurrent networks with many-to-many structure	Short term	Chou et al. (2011)
	BP	Central urban area	Have good function approximation capabilities	Short term	Long et al. (2010)
	CNN-LSTM	Urban shared parking, university parking lot	Determine spatio-temporal dependence	Short term	Liu et al. (2022)
	STGCN	Urban shared parking	Indicate spatio-temporal dependence	Short term	Xiao et al. (2021b)
	XGBoost-DT	Urban shared parking	Fit nonlinear features	Short term	Han (2022)
	GRU-GCN	Central urban area	Extract time-space information	Short term	Li and Zhong (2023)
	ICEEMDAN-GRU	Central urban area	Reduce sequence complexity and non-linearity	Short term	Li and Zhong (2023)

parking behavior and the subsequent demand for parking spaces. As the field evolves, models continue to advance in complexity, reflecting the intricate web of factors that collectively shape parking behavior and demand.

5.3. Parking space prediction

Parking space prediction is a method that forecasts the usage and availability of parking spaces within a parking lot or area based on historical data, vehicular flow, and parking facility information (Caicedo et al., 2012). Throughout this process, historical parking data, vehicular flow data, and parking layout play pivotal roles as crucial data sources (Vlahogianni et al., 2016). With the continuous advancement of technology and the promotion of data sharing, acquiring the necessary data for parking space prediction has become more convenient through means such as geomagnetic sensors, cameras, and parking management systems (Al-Turjman and Malekloo, 2019). During the construction of prediction models, factors including temporal features, vehicular flow, parking facility characteristics, weather data, socio-economic factors, events, and holidays are regarded as essential influencing elements. The ongoing improvement in feature engineering and selection methods empowers models to accurately extract and choose critical features from raw data, thus enhancing predictive performance. Regarding model development, the progress in machine learning and deep learning technologies has endowed parking space prediction models with enhanced modeling capabilities. These techniques excel in handling data intricacies and nonlinear relationships, effectively capturing patterns and trends within the data, thereby elevating prediction accuracy and reliability (Tekouabou et al., 2022).

On the basis of existing researchs, parking space prediction can be divided into micro-simulation and data-driven methods. Microsimulation methods are a modeling approach grounded in the principles of traffic flow and vehicle behavior. By constructing simulation models for vehicle movement and parking, this approach effectively simulates the motion of vehicles within parking lots, enabling predictions of parking space occupancy (Rodríguez et al., 2022). On the other hand, data-driven methods rely on historical data and encompass statistical and machine learning techniques. These methods analyze and extract patterns and regularities from historical parking data to forecast future parking space occupancy. However, microsimulation methods encounter challenges during application. One such challenge involves the collection and processing of substantial vehicular movement data and parking lot layout information. Furthermore, parameter adjustment and calibration are necessary based on the structural and traffic conditions of different parking lots to ensure model accuracy and reliability. This paper predominantly focuses on a review of data-driven approaches to parking space prediction.

Dunning (2006) proposed the development of a parking guidance and information system (PGIS) for the collection of parking data and the prediction of parking space availability. The system employed the ARIMA model to forecast parking occupancy. The study's findings revealed that the model exhibited instability and was only effective in predicting

periods of relatively stable occupancy. Its predictive accuracy was notably lower during peak or off-peak periods. Caliskan et al. (2007) utilized historical data related to vehicle arrival times and car park occupancy, collected through the PGIS, to predict parking occupancy. They employed continuous time homogeneous Markov and queuing theory prediction methods. For forecasting parking availability, Tilahun and Di Marzo Serugendo (2017) proposed the utilization of a time-varying dynamic Markov chain. The researchers constructed a transition matrix using historical data on time intervals, workdays, seasons, and weather conditions to depict long-term variations in demand. Rajabioun and Ioannou (2015) developed a multivariate autoregressive model that considered both temporal and spatial relationships among available parking spaces. They applied this model to actual data to predict arrival times and offer parking location recommendations to drivers. In a similar vein, Yu et al. (2015) employed the ARIMA model to predict the availability of vacant parking spaces. Traditional time series models used for parking space prediction demonstrate their effectiveness in this field. However, in the face of more complex parking scenarios, further research and improvements are required to enhance the accuracy of these models.

Mei et al. (2020) employed a combined approach of Fourier transform (FT) and the machine learning method least squares support vector regression (LSSVR) to validate the model's exceptional performance. The study demonstrated higher prediction precision than the traditional LSSVR model across two parking lots. Ye et al. (2020) predicted available parking spaces using gradient boosted decision trees (GBDT) and wavelet neural networks (WNN). Zheng et al. (2015) utilized three different feature combinations along with three machine learning methods, regression tree (RT), neural network (NN), and SVM, to predict parking availability, with RT emerging as the most effective. Zhao and Zhang (2020) established a parking occupancy prediction model employing four models: linear regression, SVM, BPNN, and ARIMA. The study found SVM to offer stable and reliable predictive capabilities for various car park types and sizes. Fan et al. (2018) introduced an SVM-based model for predicting the availability of unoccupied parking spaces. They incorporated the drosophila optimization algorithm to optimize the penalty factor. Tamrazian et al. (2015) employed clustering to predict parking occupancy separately for offline and online types. Awan et al. (2020) conducted a comparative analysis of novel machine learning methods and found DT to best address the challenge of predicting parking availability, closely followed by integrated learning. The study revealed that the simplicity of the k-nearest neighbor (KNN) algorithm often outperformed complex methods like multilayer perceptron. Paidi (2022) selected three methods for short-term parking availability forecasting, with results indicating that the integrated approach of DT and random forest (RF) offered accurate forecasts on both weekdays and weekends. Provoost et al. (2020) developed a machine learning prediction model utilizing NN and RF for predicting parking availability within a 60-min timeframe. These studies demonstrate the application of machine learning techniques in the field of parking prediction, revealing greater flexibility and

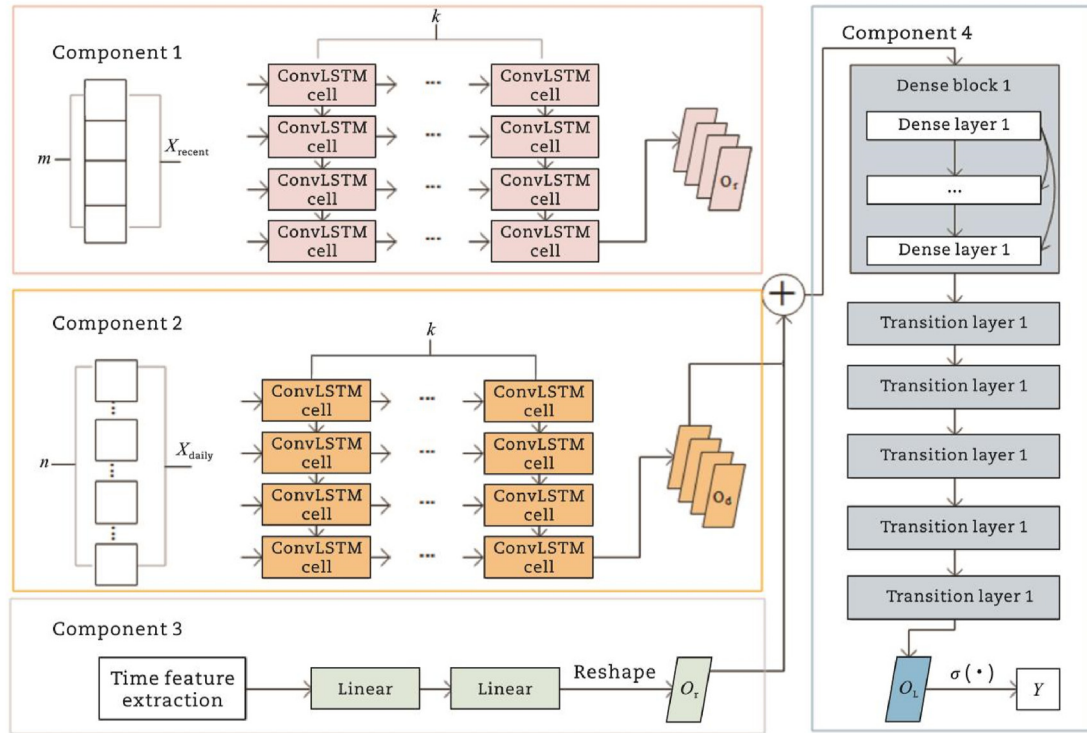


Fig. 9 – Framework of the dConvLSTM-DCN model (Feng et al., 2022a).

scalability of the models. The remarkable capability of non-linear relational modelling makes it a powerful tool for solving parking prediction challenges.

Ji et al. (2015) utilized WNN for predicting available parking spaces, showcasing superior accuracy compared to the maximum Lyapunov exponent (LE) method. Zhao et al. (2019) introduced a short-term free parking space prediction system combining multitasking with a DBN neural network. The DBN neural network was employed to analyze autocorrelation within initial data. Han et al. (2017) proposed a grey-WNN model that effectively reduced random wave-type interference in effective berth time series, rendering it more stable and accurate than traditional neural networks. Taking into account the distinctive fluctuations in effective parking berths, He et al. (2012) applied a chaotic time series method to reconstruct the phase space of historical parking berth data. They then predicted the trend of effective parking berths using a BPNN model. Chen et al. (2007) merged phase-space reconstruction with an elman neural network to achieve prediction effectiveness. Their approach demonstrated satisfactory results in short-term free parking berth prediction within parking guidance systems. Yang et al. (2019) integrated GCN and LSTM for a 30-min advance forecast of parking occupancy. The model gathered inputs from various sources including traffic, parking, and weather, surpassing multi-layer LSTM and LASSO methods. Fan et al. (2020) proposed a novel multi-step long short-term memory recurrent neural network (LSTM-NN) model, excelling in both short and long-term parking availability prediction. It

outperformed several traditional machine learning models like BPNN, SVR, stacked autoencoder (SAE), and GRU neural network models.

Feng et al. (2021, 2022b) introduced the ConvLSTM-DN, dConvLSTM-DCN, and ST-GBGRU models for short and long-term forecasting of vacant parking spaces. A graphical representation of the dConvLSTM-DCN model's framework is provided in Fig. 9. Zeng et al. (2022a) combined wavelet transform (WT) with bidirectional LSTM (BiLSTM) for denoising historical statistics, further training, and time series prediction. Liu et al. (2019) implemented prediction using LSTM and BiLSTM. Proposing a hybrid approach, Zeng et al. (2022b) stacked GRU and LSTM for parking occupancy prediction. Inputs like occupancy rate, weather conditions, and holidays were utilized to predict parking availability. Arjona et al. (2020) employed LSTM and GRU for accurate urban parking space prediction, incorporating hourly weather, calendar effects, and exogenous variables to enhance model quality. Xiao et al. (2021a) developed a deep learning framework, the hybrid spatiotemporal graph convolutional network (HST-GCN), for roadside parking availability prediction. A time-gated convolution block captured time features, while GCN mined global spatial context features. Zhang et al. (2020b) proposed a semi-supervised hierarchical recurrent graph neural network (SHARE) for predicting overall city parking space availability. RNN merged time dynamics, while context graph convolution blocks and soft clustering graph convolution blocks captured regional and global spatial relationships.

Table 5 – Classification and characteristics of deep learning-based parking prediction methods.

Model classification	Specific model	Characteristics	Prediction duration	Prediction type	Literature
Single model	WNN	Self-learning capability and fault tolerance with low computational cost	Short-term	Available parking space prediction	Ji et al. (2015)
	CNN	Accurate recognition of images	Short-term	Vacant parking space prediction	Gkolias and Vlahogianni (2018)
	LSTM	Modelling the temporal dependence of data	Short-term	Parking occupancy prediction	Canli and Toklu (2011)
	DBN	Effective use of hidden layers and robust classification	Short-term	Available parking space prediction	Zhao et al. (2019)
Hybrid model	Gray-WNN	Reduces interference with initial data volatility	Short-term	Available parking space prediction	Han et al. (2017)
	GCNN-LSTM	Modelling the spatial and temporal patterns of parking occupancy in a large road network	Short-term	Parking occupancy prediction	Yang et al. (2019)
	LSTM-NN	Optimize key model parameters	Short-term, long-term	Vacant parking space prediction	Fan et al. (2020)
	dConvLSTM-DCN	Capture the temporal and spatial characteristics of historical information and enhance the dissemination and utilization of characteristics in the prediction process	Short-term, long-term	Vacant parking space prediction	Feng et al. (2022b)
	WT-BiLSTM	Learn from historical denoising to capture long-range dependencies of time series	Short-term	Parking availability prediction	Zeng et al. (2022a)
	LSTM-BiLSTM	Enhance long-term learning ability of time series	Short-term	Parking prediction	Liu et al. (2019)
	GRU-LSTM	Improve the efficiency of model operation	Short-term	Parking occupancy, availability prediction	Arjona et al. (2020); Zeng et al. (2022b)
	HST-GCN	Explore the temporal and spatial characteristics of parking duration distribution	Short-term	Parking availability prediction	Xiao et al. (2021a)
	SHARE-RNN	Mine regional and global spatial relationships between carparks and incorporate temporal dynamics	Short-term	Parking availability prediction	Zhang et al. (2020b)
	G ² RNN	Extract both spatial and temporal correlations concurrently between different parking lots	Short-term, long-term	Parking availability prediction	Feng et al. (2023)

Feng et al. (2023) introduced a novel deep gated graph recurrent neural network (G²RNN) model, which demonstrates remarkable accuracy in both short-term and long-term predictions. The existing deep learning parking prediction models were categorized into single and combined models, as shown in Table 5. By combining diverse algorithms and features, these models exhibit enhanced adaptability and predictive performance, aiming to cater to various types and scales of parking lots. The evolution from single models to hybrid models empowers parking prediction models to better tackle the challenges and complexities present in real-world scenarios.

In conclusion, parking space prediction models are continually evolving towards a more comprehensive and holistic direction, progressively incorporating an increasing number of factors and data sources. To further enhance prediction accuracy and adapt to the increasingly complex parking environments, hybrid models have garnered the attention of researchers. By amalgamating and integrating multiple models, hybrid approaches have achieved notable

success in reconciling predictions from different models. This pivotal step has significantly contributed to enhancing prediction accuracy and robustness.

5.4. Discussion

By comprehensively reviewing the research achievements in parking behavior prediction, parking demand prediction, and parking space prediction, we can draw the following conclusions.

- (1) Basic factors such as parking location, driver characteristics, parking rate, parking time and walking distance lead to the variability of parking behavior. With the development of intelligent times, automatic driving has also become a new influencing factor. For the prediction of parking behavior, more researchers are taking into account weather and parking pattern periodicity as important features in their models to improve prediction accuracy. Anomalous events such as the

coronavirus pandemic (COVID-19) also pose prediction challenges. LSTM models with event mechanisms can achieve significant prediction results in complex situations with long prediction time horizons and limited training data. Working with a powerful smart parking platform to predict behavior on large amounts of real parking data allows for more accurate evaluation of the model. In addition, some scholars have also focused on the prediction of parking payment behavior, which can facilitate the construction of shared parking, with the latest proposed time-relative graph convolutional network achieving a prediction accuracy of 91.2%.

- (2) Researchers have made numerous parking demand forecasting methods throughout the last few decades, which have experienced the development process from land to individual behavior, from traditional regression to multidisciplinary forecasting. The historical demand data is not well adaptive and self-learning ability, can not meet the practical needs. The progress of modern transportation technology has broadened the research opportunities for the application of big data in parking demand prediction. Machine learning models based on neural networks can extract the relationship among factors such as economy, land, and traffic characteristics by virtue of their superior internal performance. Among them, CNN and LSTM are the most widely used models, which can significantly increase the stability and accuracy of prediction.
- (3) For the prediction of parking spaces, the research scholars mainly focus on three perspectives of parking availability prediction, parking occupancy prediction and vacant parking space prediction. Traditional methods and deep learning methods are employed to categorize the prediction techniques in this study. Traditional models include time series models and machine learning models. Owing to parking data's dynamic and complex nature, models are gradually changing from single models to combined models and from statistical models to deep models. The input of data has also shifted from single source data input to multiple source data input. Weather conditions, traffic, holidays and other multiple factors are integrated into the model, and LSTM-NN, ConvLSTM-DN, ST-GBGRU, LSTM-BiLSTM, and other hybrid models are emerging, which are committed to comprehensively capturing spatio-temporal correlation features and improving prediction performance. In addition, some researchers have optimized network parameters applying optimization algorithms such as whale algorithms and genetic algorithm. The continuous model optimization makes the uncertainty and the non-linear characteristics between parking data more learnable and the results of parking prediction closer to reality.

5.5. Future work

The utilization of big data in urban vehicle parking is underpinned by its inherent advantages of high frequency and steadfast demand. Predicting parking behavior, demand,

available parking spaces, and other aspects offers not only appeal to drivers and operators, but also holds critical significance for the advancement of intelligent parking, visualization of management, and efficient operational development. This, in turn, equips city administrators with pragmatic solutions to address parking challenges and instigate transformative shifts in vehicular parking and travel patterns. As such, delving deep into the intricacies of big data to uncover underlying patterns and applying them effectively in parking prediction emerges as a pivotal research trajectory. Precise predictions of parking occupancy and availability stand as cornerstones for both parking guidance systems and the implementation of sharing policies. Exploring various models to tackle this pivotal aspect has become a focal point of research interest. Given the introduction of new transportation policies and the proliferation of innovative technologies, the present era witnesses the emergence of more sophisticated forecasting models. Future research could be furthered along the following dimensions, as viewed from our vantage point.

- (1) Multivariate analysis is an intricate and intriguing domain that explores potential relationships, patterns, and correlations among diverse attributes. In the context of parking-related forecasting models, whether for predicting parking behavior or forecasting parking space availability, there is a heightened emphasis on designing the model architecture to incorporate various exceptional elements such as traffic variations, holidays, planned events, and weather conditions. The integration of these atypical factors can significantly enhance predictive accuracy. Furthermore, the purview of parking management and operations encompasses additional factors that exert influence on parking prediction. These factors encompass a wide array of considerations, including shared parking mechanisms, the influence of parking pricing on demand patterns, and the efficacy of parking reservation strategies. Integrating these aspects is vital for a comprehensive analysis and prediction of parking-related issues. Additionally, it is noteworthy that the fluctuations in parking space availability do not follow a consistent trend throughout the day, given the dynamic nature of parking data. Consequently, prediction techniques should be tailored to model the fluctuations in the number of available parking spaces across different time intervals.
- (2) Deep learning models have gained widespread adoption in parking prediction, however, these models require a substantial volume of data for effective training. Unfortunately, the collection of comprehensive parking data often faces limitations due to factors such as high costs and non-disclosure concerns. In light of these data limitations, future research endeavors should focus on developing prediction models tailored to smaller-scale datasets that can still capture spatial and temporal patterns effectively. Moreover, within the prediction process, parking data are typically fed indiscriminately into the model, neglecting the dynamic nature of parking behaviors and thereby diminishing the potential to uncover and recognize underlying parking patterns.

Additionally, the collected data is susceptible to noise interference. Moving forward, beyond the construction of reinforcement learning and noise models, there exists an opportunity to explore novel robust statistical methodologies, such as robust regression and anomaly detection. These approaches can be employed to bolster parking prediction capabilities, particularly in environments characterized by substantial noise. Furthermore, there is promise in considering the fusion of information from diverse data sources to counteract noise interference stemming from a single data source. By pursuing these avenues, the potential exists to enhance the precision and resilience of parking prediction, offering prospects for advancements in the field.

- (3) Parking prediction models are progressively transitioning from single models to integrated models and from statistical approaches to deep learning methodologies. While deep learning models represent an advancement over traditional machine learning models, their intricate and sophisticated architectures can pose challenges in practical applications. The process of stacking different models and levels within neural networks can significantly extend the training duration of these models. Addressing this concern, there is merit in exploring methods to simplify deep architectures, making them more accessible for practical implementation. The emergence of data-driven technologies has driven the need for advanced big data techniques capable of handling vast volumes of parking-related data. Within the realm of big data paradigms (e.g., parallel computing, cloud computing), deep learning-based parking prediction models have the potential to yield efficient and cost-effective predictions. Furthermore, the incorporation of unstructured datasets poses a unique challenge in this domain. Finding effective ways to incorporate and analyze unstructured data is a frontier that warrants exploration.
- (4) As an innovative transportation paradigm, an increasing focus is being directed towards shared parking lots as research subjects. The benefits brought about by parking sharing systems, which contribute to alleviating urban congestion and enhancing the quality of life and economic vitality in cities, play a crucial role in curbing parking demand. Presently, existing research predominantly addresses short-term prediction within the context of shared parking lots. However, the long-term demand forecast holds paramount importance for achieving equilibrium between parking supply and demand, as well as ensuring the sustainable progression of shared parking systems.
- (5) In contemporary times, the majority of modern parking facilities incorporate Internet of Things (IoT) technologies, such as license plate recognition, automatic access control, and mobile payment systems, to enhance the convenience for both parking lot operators and drivers. The intelligent utilization of parking-related big data is becoming more prevalent, leading to substantial enhancements in the performance of intelligent parking systems. Building upon the aforementioned analysis, several studies have been integrated with extensive

parking management platforms or parking applications (apps). The effective harnessing of big data for practical and tangible projects, aided by predictive models, to address parking challenges presents an additional noteworthy challenge.

6. Conclusions

Parking plays a crucial role in residents' transportation choices, and the existing challenges of parking difficulties have introduced new complexities. These challenges encompass rigid parking demands, limited land resources, temporal and spatial imbalances in parking supply and demand, as well as underutilized parking spaces, all of which pose considerable hurdles for parking operators and city administrators. In response to the escalating parking predicament, numerous scholars have focused on alleviating parking pressures, yielding fruitful research outcomes. Within this realm, parking prediction has emerged as a pivotal technology and effective approach to mitigate the parking problem. Diverse prediction models have also proliferated to address this issue.

To comprehensively understand the relevant research in parking prediction. Firstly, we conducted a comprehensive statistical analysis of the development and partnership between authors, institutions, and countries. Simultaneously, we revealed the correlation between keywords and the evolution process of topics. This provides valuable references for future research and development in the field of parking prediction. Furthermore, we categorized the advancements in parking prediction models, encompassing time series models, machine learning models, and deep learning models. For each model category, we provided detailed summaries of their unique characteristics, advantages, and limitations. Finally, we systematically classified three major hot topics in parking prediction: parking behavior prediction, parking demand prediction, and parking space prediction. We provided a comprehensive overview of the applications of various models in these problems and discussed the potential challenges and directions for future research.

Conflict of interest

The authors do not have any conflict of interest with other entities or researchers.

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